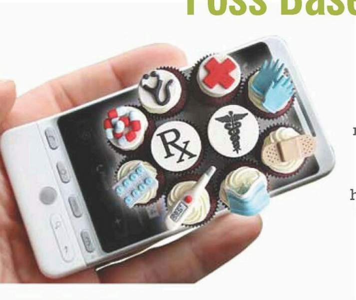
Foss Based Mobile Solutions
Deliver Medical Facilities to Remote Areas



In the last several years, across the globe, the number of mobile phone users has exploded. The healthcare industry has realised their untapped potential, and frequently uses crowd-sourcing and crowd-mapping on mobiles for post-disaster healthcare support planning. This article, intended for doctors, healthcare systems service providers or those interested in the benefits of open source applications for public healthcare infrastructure, explores software platforms like SMS for Life, Ushahidi and SwiftRiver.

Mobiles are becoming more and more important in public health. Earlier, Google and Frontline SMS collaborated during the Haiti post-earthquake disaster-relief efforts, and used mobile technology to make healthcare workers aware of public health issues on the ground. There was also Ushahidi to help translate text messages and tweets to data on a map.

So, what is crowd-mapping? For post-disaster relief or during any public-health crisis, anyone can send a message related to public health to an SMS short code, or send a tweet containing relevant information along with a location. This can then be translated to data on a map, allowing public-health workers to efficiently allocate resources in real-time. These real-time online interactive maps are called crowd-mapping. Once this information is collected, volunteers collaborate over the Internet to put each post onto a map, which is updated second by second, and enables officials on the ground to quickly locate people with the most critical needs, and efficiently deliver medical and humanitarian relief to them.

SMS for Life

First-line malaria medication needs to be administered within the first 24 to 48 hours after symptoms show up, in order to save the lives of children under five, and stock run-outs are such a common problem that up-to-date information on supply levels is critical. SMS for Life is a project that tracks the supply levels of malaria medication in hospitals and clinics in Africa. Health workers send weekly text messages to report on their supply levels, and the data is collected and displayed on a map. District medical officers can then view the up-to-date status of facilities in their district, and use the information to make forecasts and plan shipments. The SMS for Life Map Viewer is open source and released under the Apache 2.0 licence. It is a solution to a long-standing problem.

The SMS for Life pilot was launched on September 21, 2009, in Tanzania, in the three districts of Lindi Rural, Ulanga and Kigoma, at 155 public health facilities. It ran for five months and ended in February 2010. The pilot provided clear visibility of anti-malarial drug stock levels and hence supported more efficient stock management using simple and widely available SMS technology. This was done via a public-private partnership model that worked very effectively. The SMS for Life system has the potential to ensure the availability of anti-malarial drugs or other medicines in rural or under-resourced areas. Overall, the system was built to be a generic and highly scalable solution that can be leveraged to support any medicine or product, and can be implemented in any country with minimal tailoring. Additionally, the system could also be utilised for disease surveillance. The data captured through the SMS stock-count messages is available through a secure reporting website. The website can then be accessed via the Internet on a computer, Blackberry or other smart phone. And access to the website is granted through unique user IDs and passwords allocated at the group level.

SMS for Life automatically sends text messages to all health facilities on a weekly basis, asking for their current stock of medicines. The collected responses, stored centrally on a website, will generate the following types of reports:

- Display of the stock situation at health facilities, at the country and district levels.
- Alerts sent to the district or central stores will provide immediate information on stock run-outs.
- Query and reporting of stocks by the health facility, district, region and zone, as well as the calculated average weekly usage by each health facility, etc.
- Early warning of malaria outbreaks, e.g., highlighting when there is a sharp rise in weekly usage in a number of closely located health facilities.

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